13

The Average Causal Effect on the Treated Units and Other Estimands

Chapters 10–12 focused on the identification and estimation of the average causal effect $\tau = E\{Y(1) - Y(0)\}$ under the ignorability and overlap assumptions. Conceptually, it is straightforward to extend the discussion to the average causal effects on the treated and control units:

$$\begin{aligned} \tau_{\mathrm{T}} &= E\{Y(1) - Y(0) \mid Z = 1\}, \\ \tau_{\mathrm{C}} &= E\{Y(1) - Y(0) \mid Z = 0\}. \end{aligned}$$

If τ_{T} and τ_{C} differ from τ, then the average causal effects are heterogeneous across the treatment and control groups. Whether we should estimate τ_{T}, τ_{C} or τ depends on the practical question of interest.

Because of the symmetry, this chapter focuses on τ_{T}. Chapter 13.4 also discusses extensions to other estimands.

13.1 Nonparametric identification of τ_{T}

The average causal effect on the treated units equals

$$\tau_{\mathrm{T}} = E(Y \mid Z = 1) - E\{Y(0) \mid Z = 1\},$$

where the first term $E(Y \mid Z = 1)$ is directly identifiable from the data and the second term $E\{Y(0) \mid Z = 1\}$ is counterfactual. The key assumption to identify the second term is the following ignorability and overlap assumptions.

Assumption 13.1 *$Z \perp\!\!\!\perp Y(0) \mid X$ and $e(X) < 1$.*

Because the key is to identify $E\{Y(0) \mid Z = 1\}$, we only need the "one-sided" ignorability and overlap assumptions. Under Assumption 13.1, we have the following identification result for τ_{T}.

Theorem 13.1 *Under Assumption 13.1, we have*

$$\begin{aligned} E\{Y(0) \mid Z = 1\} &= E\left\{E(Y \mid Z = 0, X) \mid Z = 1\right\} \\ &= \int E(Y \mid Z = 0, X = x) f(x \mid Z = 1) \mathrm{d}x. \end{aligned}$$

By Theorem 13.1 the counterfactual mean $E\{Y(0) \mid Z=1\}$ equals the conditional mean of the observed outcomes under the control, averaged over the distribution of the covariates under the treatment. It implies that τ_{T} is nonparametrically identified by

$$\tau_{\mathrm{T}} = E(Y \mid Z=1) - E\{E(Y \mid Z=0, X) \mid Z=1\} \tag{13.1}$$

Proof of Theorem 13.1: We have

$$\begin{aligned} E\{Y(0) \mid Z=1\} &= E\big[E\{Y(0) \mid Z=1, X\} \mid Z=1\big] \\ &= E\big[E\{Y(0) \mid Z=0, X\} \mid Z=1\big] \\ &= E\{E(Y \mid Z=0, X) \mid Z=1\} \\ &= \int E(Y \mid Z=0, X=x) f(x \mid Z=1)\mathrm{d}x. \end{aligned}$$

$\square$

With a discrete X, the identification formula in Theorem 13.1 reduces to

$$E\{Y(0) \mid Z=1\} = \sum_{k=1}^{K} E(Y \mid Z=0, X=k)\mathrm{pr}(X=k \mid Z=1),$$

motivating the following stratified estimator for τ_{T}:

$$\hat{\tau}_{\mathrm{T}} = \hat{\bar{Y}}(1) - \sum_{k=1}^{K} \hat{\pi}_{[k]|1} \hat{\bar{Y}}_{[k]}(0),$$

where $\hat{\pi}_{[k]|1} = n_{[k]1}/n_1$ is the proportion of category k of X among the treated units.

For continuous X, we need to fit an outcome model for $E(Y \mid Z=0, X)$ using the control units. If the fitted values for the control potential outcomes are $\hat{\mu}_0(X_i)$, then the outcome regression estimator is

$$\hat{\tau}_{\mathrm{T}}^{\mathrm{reg}} = \hat{\bar{Y}}(1) - n_1^{-1}\sum_{i=1}^{n} Z_i \hat{\mu}_0(X_i) = n_1^{-1}\sum_{i=1}^{n} Z_i\{Y_i - \hat{\mu}_0(X_i)\}.$$

Example 13.1 *If we specify a linear model for all units*

$$E(Y \mid Z, X) = \beta_0 + \beta_z Z + \beta_x^{\mathsf{T}} X,$$

then

$$\begin{aligned} \tau_{\mathrm{T}} &= E\{E(Y \mid Z=1, X) - E(Y \mid Z=0, X) \mid Z=1\} \\ &= \beta_z. \end{aligned}$$

If we run OLS to obtain $(\hat{\beta}_0, \hat{\beta}_z, \hat{\beta}_x)$, then we can use $\hat{\beta}_z$ to estimate τ_{T}. Section 10.4.2 shows that $\hat{\beta}_z$ is an estimator for τ, and this example further shows that $\hat{\beta}_z$ is an estimator for τ_{T}. This is not surprising because the linear model assumes constant causal effects across units.

Example 13.2 *The identification formula depends only on $E(Y \mid Z=0,X)$, so we need only to specify a model for the control units. When this model is linear,*

$$E(Y \mid Z=0,X)=\beta_{0|0}+\beta_{x|0}^{\mathsf{T}}X,$$

we have

$$\begin{aligned}\tau_{\mathrm{T}} &= E(Y \mid Z=1)-E(\beta_{0|0}+\beta_{x|0}^{\mathsf{T}}X \mid Z=1)\\ &= E(Y \mid Z=1)-\beta_{0|0}-\beta_{x|0}^{\mathsf{T}}E(X \mid Z=1).\end{aligned}$$

If we run OLS with only the control units to obtain $(\hat\beta_{0|0},\hat\beta_{x|0})$, then the estimator is

$$\hat\tau_{\mathrm{T}}=\hat{\bar{Y}}(1)-\hat\beta_{0|0}-\hat\beta_{x|0}^{\mathsf{T}}\hat{\bar{X}}(1).$$

Using the property of the OLS (see B.5), we have

$$\hat{\bar{Y}}(0)=\hat\beta_{0|0}+\hat\beta_{x|0}^{\mathsf{T}}\hat{\bar{X}}(0).$$

Therefore, the above estimator reduces to

$$\hat\tau_{\mathrm{T}}=\left\{\hat{\bar{Y}}(1)-\hat{\bar{Y}}(0)\right\}-\hat\beta_{x|0}^{\mathsf{T}}\left\{\hat{\bar{X}}(1)-\hat{\bar{X}}(0)\right\},$$

which is similar to (**??**) *with a different coefficient for the difference in means of the covariates.*

As an algebraic fact, we can show that this estimator equals the coefficient of Z in the OLS fit of the outcome on the treatment, covariates, and their interactions, with the covariates centered as $X_i-\hat{\bar{X}}(1)$. See Problem 13.2 for more details.

13.2 Inverse propensity score weighting and doubly robust estimation of τ_{T}

Theorem 13.2 *Under Assumption 13.1, we have*

$$E\{Y(0) \mid Z=1\}=E\left\{\frac{e(X)}{e}\frac{1-Z}{1-e(X)}Y\right\} \tag{13.2}$$

and

$$\tau_{\mathrm{T}}=E(Y \mid Z=1)-E\left\{\frac{e(X)}{e}\frac{1-Z}{1-e(X)}Y\right\}, \tag{13.3}$$

where $e=\mathrm{pr}(Z=1)$ is the marginal probability of the treatment.

Proof of Theorem 13.2: The left-hand side of (13.2) equals

$$\begin{aligned} E\{Y(0) \mid Z=1\} &= E\{ZY(0)\}/e \\ &= E\big[E(Z \mid X)E\{Y(0) \mid X\}\big]/e \\ &= E\big[e(X)E\{Y(0) \mid X\}\big]/e. \end{aligned}$$

The right-hand side of (13.2) equals

$$\begin{aligned} E\left\{\frac{e(X)}{e}\frac{1-Z}{1-e(X)}Y\right\} &= E\left[E\left\{\frac{e(X)}{e}\frac{1-Z}{1-e(X)}Y(0) \mid X\right\}\right] \\ &= E\left[\frac{e(X)}{e\{1-e(X)\}}E\left\{(1-Z)Y(0) \mid X\right\}\right] \\ &= E\left[\frac{e(X)}{e\{1-e(X)\}}E(1-Z \mid X)E\left\{Y(0) \mid X\right\}\right] \\ &= E\big[e(X)E\{Y(0) \mid X\}\big]/e. \end{aligned}$$

So (13.2) holds. □

We have two IPW estimators

$$\hat{\tau}_{\mathrm{T}}^{\mathrm{ht}} = \hat{\bar{Y}}(1) - n_1^{-1}\sum_{i=1}^{n}\hat{o}(X_i)(1-Z_i)Y_i$$

and

$$\hat{\tau}_{\mathrm{T}}^{\mathrm{hajek}} = \hat{\bar{Y}}(1) - \frac{\sum_{i=1}^{n}\hat{o}(X_i)(1-Z_i)Y_i}{\sum_{i=1}^{n}\hat{o}(X_i)(1-Z_i)},$$

where $\hat{o}(X_i) = \hat{e}(X_i)/\{1-\hat{e}(X_i)\}$ is the fitted odds of the treatment given covariates.

We also have a doubly robust estimator for $E\{Y(0) \mid Z=1\}$ which combines the propensity score and the outcome models. Define

$$\tilde{\mu}_{0\mathrm{T}}^{\mathrm{dr}} = E\left[o(X,\alpha)(1-Z)\{Y-\mu_0(X,\beta_0)\} + Z\mu_0(X,\beta_0)\right]/e, \qquad (13.4)$$

where $o(X,\alpha) = e(X,\alpha)/\{1-e(X,\alpha)\}$.

Theorem 13.3 *Under Assumption 13.1, if either* $e(X,\alpha) = e(X)$ *or* $\mu_0(X,\beta_0) = \mu_0(X)$*, then* $\mu_{0\mathrm{T}}^{dr} = E\{Y(0) \mid Z=1\}$.

Proof of Theorem 13.3: We have the decomposition

$$
\begin{aligned}
& e\left[\tilde{\mu}_{0\mathrm{T}}^{\mathrm{dr}} - E\{Y(0) \mid Z=1\}\right] \\
= \; & E\left[o(X,\alpha)(1-Z)\{Y(0)-\mu_0(X,\beta_0)\} + Z\mu_0(X,\beta_0)\right] - E\{ZY(0)\} \\
= \; & E\left[o(X,\alpha)(1-Z)\{Y(0)-\mu_0(X,\beta_0)\} - Z\{Y(0)-\mu_0(X,\beta_0)\}\right] \\
= \; & E\left[\{o(X,\alpha)(1-Z)-Z\}\{Y(0)-\mu_0(X,\beta_0)\}\right] \\
= \; & E\left[\frac{e(X,\alpha)-Z}{1-e(X,\alpha)}\{Y(0)-\mu_0(X,\beta_0)\}\right] \\
= \; & E\left[E\left\{\frac{e(X,\alpha)-Z}{1-e(X,\alpha)} \mid X\right\} \times E\{Y(0)-\mu_0(X,\beta_0) \mid X\}\right] \\
= \; & E\left[\frac{e(X,\alpha)-e(X)}{1-e(X,\alpha)} \times \{\mu_0(X)-\mu_0(X,\beta_0)\}\right].
\end{aligned}
$$

Therefore, $\tilde{\mu}_{0\mathrm{T}}^{\mathrm{dr}} - E\{Y(0) \mid Z=1\} = 0$ if either $e(X,\alpha)=e(X)$ or $\mu_0(X,\beta_0) = \mu_0(X)$. □

Based on the population versions of $\tilde{\mu}_{0\mathrm{T}}^{\mathrm{dr}}$ in (13.4), we can obtain its sample version to construct a doubly robust estimator for τ_{T}.

Definition 13.1 (doubly robust estimator for $\tau_{\mathbf{T}}$) *Based on the data $(X_i, Z_i, Y_i)_{i=1}^n$, we can obtain a doubly robust estimator for τ_{T} by the following steps:*

1.obtain the fitted values of the propensity scores $e(X_i,\hat{\alpha})$ and then obtain the fitted values of the odds $o(X_i,\hat{\alpha}) = e(X_i,\hat{\alpha})/(1 - e(X_i,\hat{\alpha}))$;

2.obtain the fitted values of the outcome mean under control $\mu_0(X_i,\hat{\beta}_0)$;

3.construct the doubly robust estimator: $\hat{\tau}_{\mathrm{T}}^{\mathrm{dr}} = \hat{\bar{Y}}(1) - \hat{\mu}_{0\mathrm{T}}^{\mathrm{dr}}$, where

$$
\hat{\mu}_{0\mathrm{T}}^{\mathrm{dr}} = \frac{1}{n_1}\sum_{i=1}^{n}\left[o(X_i,\hat{\alpha})(1-Z_i)\{Y_i-\mu_0(X_i,\hat{\beta}_0)\} + Z_i\mu_0(X_i,\hat{\beta}_0)\right].
$$

By Definition 13.1, we can rewrite $\hat{\tau}_{\mathrm{T}}^{\mathrm{dr}}$ as

$$
e(X_i,\hat{\alpha}) = \hat{\tau}_{\mathrm{T}}^{\mathrm{reg}} - \frac{1}{n_1}\sum_{i=1}^{n} o(X_i,\hat{\alpha})(1-Z_i)\{Y_i-\mu_0(X_i,\hat{\beta}_0)\}
$$

or

$$
e(X_i,\hat{\alpha}) = \hat{\tau}_{\mathrm{T}}^{\mathrm{ht}} - \frac{1}{n_1}\sum_{i=1}^{n}\{o(X_i,\hat{\alpha})(1-Z_i)+Z_i\}\mu_0(X_i,\hat{\beta}_0).
$$

Similar to the discussion of $\hat{\tau}^{\mathrm{dr}}$, we can estimate the variance of $\hat{\tau}_{\mathrm{T}}^{\mathrm{dr}}$ via the bootstrap by resampling from $(Z_i, X_i, Y_i)_{i=1}^n$. Hahn (1998), Mercatanti and Li (2014), Shinozaki and Matsuyama (2015) and Yang and Ding (2018b) are references on the estimation of τ_{T}.

13.3 An example

The following R function implements two outcome regression estimators, two IPW estimators, and the doubly robust estimator for τ_{T}. To avoid extreme estimated propensity scores, we can also truncate them from the above.

```
ATT.est = function(z, y, x, out.family = gaussian, Utruncps = 1)
{
  ## sample size
  nn  = length(z)
  nn1 = sum(z)

  ## fitted propensity score
  pscore   = glm(z ~ x, family = binomial)$fitted.values
  pscore   = pmin(Utruncps, pscore)
  odds     = pscore/(1 - pscore)

  ## fitted potential outcomes
  outcome0 = glm(y ~ x, weights = (1 - z),
                 family = out.family)$fitted.values

  ## outcome regression estimator
  ace.reg0 = lm(y ~ z + x)$coef[2]
  ace.reg  = mean(y[z==1]) - mean(outcome0[z==1])
  ## propensity score weighting estimator
  ace.ipw0 = mean(y[z==1]) -
              mean(odds*(1 - z)*y)*nn/nn1
  ace.ipw  = mean(y[z==1]) -
              mean(odds*(1 - z)*y)/mean(odds*(1 - z))
  ## doubly robust estimator
  res0     = y - outcome0
  ace.dr   = ace.reg - mean(odds*(1 - z)*res0)*nn/nn1

  return(c(ace.reg0, ace.reg, ace.ipw0, ace.ipw, ace.dr))
}
```

The following R function further implements the bootstrap variance estimators.

```
OS_ATT = function(z, y, x, n.boot = 10^2,
                  out.family = gaussian, Utruncps = 1)
{
  point.est  = ATT.est(z, y, x, out.family, Utruncps)

  ## nonparametric bootstrap
  n   = length(z)
  x          = as.matrix(x)
  boot.est   = replicate(n.boot, {
    id.boot = sample(1:n, n, replace = TRUE)
```

```
    ATT.est(z[id.boot], y[id.boot], x[id.boot, ],
            out.family, Utruncps)
  })

  boot.se     = apply(boot.est, 1, sd)

  res         = rbind(point.est, boot.se)
  rownames(res) = c("est", "se")
  colnames(res) = c("reg0", "reg", "HT", "Hajek", "DR")

  return(res)
}
```

Now we re-analyze the data in Example 10.3 to estimate τ_{T}. We obtain

```
     reg0    reg     HT  Hajek     DR
est 0.061 -0.351 -1.992 -0.351 -0.187
se  0.227  0.258  0.705  0.328  0.287
```

without truncating the estimated propensity scores, and

```
     reg0    reg     HT  Hajek     DR
est 0.061 -0.351 -0.597 -0.192 -0.230
se  0.223  0.255  0.579  0.302  0.276
```

by truncating the estimated propensity scores from the above at 0.9. The HT estimator is sensitive to the truncation as expected. The regression estimator in Example 13.1 is quite different from other estimators. It imposes an unnecessary assumption that the regression functions in the treatment and control group share the same coefficient of X. The regression estimator in Example 13.2 is much closer to the Hajek and doubly robust estimators. The estimates above are slightly different from those in Section 12.3.3, suggesting the existence of treatment effect heterogeneity across τ_{T} and τ.

13.4 Other estimands

Li et al. (2018a) gave a unified discussion of the causal estimands in observational studies. Starting from the conditional average causal effect $\tau(X)$, they proposed a general class of estimands

$$\tau^h = \frac{E\{h(X)\tau(X)\}}{E\{h(X)\}}$$

indexed by a weighting function $h(X)$ with $E\{h(X)\} \neq 0$. The normalization in the denominator is to ensure that a constant causal effect $\tau(X) = \tau$ averages to the same τ.

Under ignorability,

$$\tau^h = \frac{E[h(X)\{\mu_1(X) - \mu_0(X)\}]}{E\{h(X)\}}$$

which motivates the outcome regression estimator

$$\hat{\tau}^h = \frac{\sum_{i=1}^n h(X_i)\{\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i)\}}{\sum_{i=1}^n h(X_i)}.$$

Moreover, we can show that τ^h has the following weighting form:

Theorem 13.4 *Under the ignorability and overlap assumption, we have*

$$\tau^h = E\left\{\frac{ZYh(X)}{e(X)} - \frac{(1-Z)Yh(X)}{1-e(X)}\right\} / E\{h(X)\}.$$

The proof of Theorem 13.4 is similar to those of Theorems 11.2 and 13.2 which is relegated to Problem 13.9. Based on Theorem 13.4, we can construct the corresponding IPW estimator for τ^h.

By Theorem 13.4, each unit is associated with the weight due to the definition of the estimand as well as the weight due to the inverse of the propensity score. Finally, the treated units are weighted by $h(X)/e(X)$ and the control units are weighted by $h(X)/\{1-e(X)\}$. Li et al. (2018a, Table 1) summarized several estimands, and I present a part of it below:

population	$h(X)$	estimand	weights
combined	1	τ	$1/e(X)$ and $1/\{1-e(X)\}$
treated	$e(X)$	τ_{T}	1 and $e(X)/\{1-e(X)\}$
control	$1-e(X)$	τ_{C}	$\{1-e(X)\}/e(X)$ and 1
overlap	$e(X)\{1-e(X)\}$	τ_{O}	$1-e(X)$ and $e(X)$

The overlap population and the corresponding estimand

$$\tau_{\mathrm{O}} = \frac{E\left[e(X)\{1-e(X)\}\tau(X)\right]}{E\left[e(X)\{1-e(X)\}\right]}$$

is new to us. This estimand has the largest weight for units with $e(X) = 1/2$ and down weights the units with extreme propensity scores. A nice feature of this estimand is that its IPW estimator is stable without the possibly extremely small values of $e(X)$ and $1-e(X)$ in the denominator. If $e(X) \perp\!\!\!\perp \tau(X)$ including the special case of $\tau(X) = \tau$, the parameter τ_{O} reduces to τ. In general, however, the estimand τ_{O} may cause controversy because it changes the initial population and depends on the propensity score which may be misspecified in practice. Li et al. (2018a) and Li et al. (2019) gave some justifications and numerical evidence. This estimand will appear again in Chapter 14.

We can also construct the doubly robust estimator for τ^h. I relegate the details to Problem 13.10.

13.5 Homework Problems

13.1 Comparing $\tau_{\mathrm{T}}, \tau_{\mathrm{C}}$, and τ

Assume $Z \perp\!\!\!\perp \{Y(1), Y(0)\} \mid X$. Recall $e(X) = \mathrm{pr}(Z = 1 \mid X)$ is the propensity score, $e = \mathrm{pr}(Z = 1)$ is the marginal probability of the treatment, and $\tau(X) = E\{Y(1) - Y(0) \mid X\}$ is the CATE. Show that

$$\tau_{\mathrm{T}} - \tau = \frac{\mathrm{cov}\{e(X), \tau(X)\}}{e}, \quad \tau_{\mathrm{C}} - \tau = -\frac{\mathrm{cov}\{e(X), \tau(X)\}}{1 - e}.$$

Remark: The results above also imply that $\tau_{\mathrm{T}} - \tau_{\mathrm{C}} = \mathrm{cov}\{e(X), \tau(X)\}/\{e(1-e)\}$. So the differences among $\tau_{\mathrm{T}}, \tau_{\mathrm{C}}, \tau$ depend on the covariance between the propensity score and CATE. When $e(X)$ and $\tau(X)$ are uncorrelated, their differences are 0.

13.2 An algebraic fact about a regression estimator for τ_{T}

This problem provides more details for Example 13.2.

Show that if we center the covariates by $X_i - \hat{\bar{X}}(1)$ for all units, then $\hat{\tau}_{\mathrm{T}}$ equals the coefficient of Z in the OLS fit of the outcome on the intercept, the treatment, covariates, and their interactions.

13.3 Simulation for the average causal effect on the treated units

Chapter 12 ran some simulation studies for τ. Run similar simulation studies for τ_{T} with either correct or incorrect propensity score or outcome models.

You can choose different model parameters, a larger number of simulation settings, and a larger number of bootstrap replicates. Report your findings, including at least the bias, variance, and variance estimator via the bootstrap. You can also report other properties of the estimators, for example, the asymptotic Normality and the coverage rates of the confidence intervals.

13.4 An alternative form of the doubly robust estimator for τ_{T}

Motivated by (13.4), we have an alternative form of doubly robust estimator for $E\{Y(0) \mid Z = 1\}$:

$$\tilde{\mu}_{0\mathrm{T}}^{\mathrm{dr2}} = \frac{E\left[o(X, \alpha)(1 - Z)\{Y - \mu_0(X, \beta_0)\}\right]}{E\left[o(X, \alpha)(1 - Z)\right]} + E\{Z\mu_0(X, \beta_0)\}/e.$$

Show that under Assumption 13.1, $\tilde{\mu}_{0\mathrm{T}}^{\mathrm{dr2}} = E\{Y(0) \mid Z = 1\}$ if either $e(X, \alpha) = e(X)$ or $\mu_0(X, \beta_0) = \mu_0(X)$. Give the sample analog of the doubly robust estimator for τ_{T}.

13.5 Average causal effect on the control units

Prove the identification formulas for τ_{C}, analogous to (13.1) and (13.3). Propose the doubly robust estimator for τ_{C}.

13.6 Estimating individual effect and conditional average causal effect

Assume that $\{Z_i, X_i, Y_i(1), Y_i(0)\}_{i=1}^n \overset{\mathrm{IID}}{\sim} \{Z, X, Y(1), Y(0)\}$. The individual effect is $\tau_i = Y_i(1) - Y_i(0)$ and the conditional average causal effect is $\tau(X_i) = E\{Y_i(1) - Y_i(0) \mid X_i\}$. Since we will discuss the individual effect, we do not drop the subscript i since τ means the average causal effect, not the population version of $Y(1) - Y(0)$.

1. Under randomization with $Z_i \perp\!\!\!\perp \{Y_i(1), Y_i(0)\}$ and $e = \mathrm{pr}(Z_i = 1)$, show that
$$\delta_i = \frac{Z_i Y_i}{e} - \frac{(1 - Z_i) Y_i}{1 - e}$$
is an unbiased predictor of the individual effect in the sense that
$$E(\delta_i - \tau_i) = 0 \quad (i = 1, \ldots, n).$$
Further show that $E(\delta_i) = \tau$ for all $i = 1, \ldots, n$.
2. Under ignorability with $Z_i \perp\!\!\!\perp \{Y_i(1), Y_i(0)\} \mid X_i$ and $e(X_i) = \mathrm{pr}(Z_i = 1 \mid X_i)$, show that
$$\delta_i = \frac{Z_i Y_i}{e(X_i)} - \frac{(1 - Z_i) Y_i}{1 - e(X_i)}$$
is an unbiased predictor of the individual effect and the conditional average causal effect in the sense that
$$E(\delta_i - \tau_i) = 0, \quad E\{\delta_i - \tau(X_i)\} = 0, \quad (i = 1, \ldots, n).$$
Further show that $E(\delta_i) = \tau$ for all $i = 1, \ldots, n$.

13.7 General estimand and $(\tau_{\mathrm{T}}, \tau_{\mathrm{C}})$

Assume unconfoundedness. Show that $\tau^h = \tau_{\mathrm{T}}$ if $h(X) = e(X)$, and $\tau^h = \tau_{\mathrm{C}}$ if $h(X) = 1 - e(X)$.

13.8 More on τ_{O}

Show that
$$\tau_{\mathrm{O}} = \frac{E[\{1 - e(X)\}\tau(X) \mid Z = 1]}{E\{1 - e(X) \mid Z = 1\}} = \frac{E\{e(X)\tau(X) \mid Z = 0\}}{E\{e(X) \mid Z = 0\}}.$$

13.9 IPW for the general estimand

Prove Theorem 13.4.

13.10 Doubly robust estimation for general estimand

For a given $h(X)$, we have the following formulas for constructing the doubly robust estimator for τ^h:

$$
\begin{aligned}
\tilde{\mu}_1^{h,\mathrm{dr}} &= E\left[\frac{Zh(X)\{Y-\mu_1(X,\beta_1)\}}{e(X,\alpha)} + h(X)\mu_1(X,\beta_1)\right], \\
\tilde{\mu}_0^{h,\mathrm{dr}} &= E\left[\frac{(1-Z)h(X)\{Y-\mu_0(X,\beta_0)\}}{1-e(X,\alpha)} + h(X)\mu_0(X,\beta_0)\right].
\end{aligned}
$$

Show that under ignorability and overlap,

1. if either $e(X,\alpha) = e(X)$ or $\mu_1(X,\beta_1) = \mu_1(X)$, then $\tilde{\mu}_1^{h,\mathrm{dr}} = E\{h(X)Y(1)\}$;
2. if either $e(X,\alpha) = e(X)$ or $\mu_0(X,\beta_0) = \mu_0(X)$, then $\tilde{\mu}_0^{h,\mathrm{dr}} = E\{h(X)Y(0)\}$;
3. if either $e(X,\alpha) = e(X)$ or $\{\mu_1(X,\beta_1) = \mu_1(X), \mu_0(X,\beta_0) = \mu_0(X)\}$, then
$$\frac{\tilde{\mu}_1^{h,\mathrm{dr}} - \tilde{\mu}_0^{h,\mathrm{dr}}}{E\{h(X)\}} = \tau^h.$$

Remark: Tao and Fu (2019) proved the above results. However, they hold only for a given $h(X)$. The most interesting cases of $\tau_{\mathrm{T}}, \tau_{\mathrm{C}}$ and τ_{O} all have weight depending on the propensity score $e(X)$, which must be estimated in the first place. The above formulas do not apply to constructing the doubly robust estimators for τ_{T} and τ_{C}; there does not exist a doubly robust estimator for τ_{O}.

13.11 Analyzing a dataset from the Karolinska Institute

Revisit Problem 12.5. Estimate τ_{T} based on the methods introduced in this Chapter.

13.12 Recommended reading

Shinozaki and Matsuyama (2015) focused on τ_{T}, and Li et al. (2018a) discussed general τ^h.